Unit-5 LASERS

(Topics)

- Properties of Laser, Einstein's theory of matter radiation: A and B coefficients,
- Amplification of light by population inversion,
- Different types of lasers,
- Gas lasers (He-Ne),
- Solid-state lasers (Ruby laser),
- Properties of laser beams: Mono-chromaticity, coherence, directionality and brightness, laser speckles,
- Applications of lasers in science, engineering, and medicine

INTRODUCTION

- LASER is an acronym for **L**ight **A**mplification by **S**timulated **E**mission of **R**adiation.
- In laser, the intensity of light is amplified by the process called stimulated emission.
- Lasers are optical phenomena that find major applications in various fields such as medicine, engineering, fiber optic communication, industries, etc.
- Light emitted by Laser is more powerful than ordinary light source.

Sr. no.	Questions
1.	**Write full form of LASER and Discuss in brief the characteristics of laser light.** **Or** **Explain various properties of the LASER beam.** **Sol:** Laser light is highly powerful, is capable of propagating over long distances, and is not easily absorbed by water. The following characteristics distinguish a laser beam from an ordinary light. **1 Coherence** • The wave trains which are identical in phase and direction are called coherent waves. • Since all the constituent photons of laser beam possess the same energy, and momentum and propagate in same direction, the laser beam is said to be highly coherent. **2 High Intensity** Due to the coherent nature of laser, it has the ability to focus over a small area of 10^{-6} cm^2, i.e., an extremely high concentration of its energy over a small area.

3 High Directionality

- An ordinary light source emits light in all possible directions. But, since laser travels as a parallel beam it can travel over a long distance without spreading.
- The angular spread of a laser beam is Imm/meter. This reveals the directionality of the laser beam.

4 High Monochromaticity

- The light from a normal monochromatic source spreads over a range of wavelengths of the order 100 nm. But, the spread is of the order of 1 nm for laser.
- Hence, a laser is highly monochromatic, i.e., it can emit light of single wavelength.

5 Laser Speckles

- Laser speckle was first observed in the 1960s, when laser light hits a surface with imperfections, it creates a unique pattern of scattered light intensity. This phenomenon leads to the creation of an interference pattern known as **speckles**.
- These speckles comprise a pattern characterized by alternating bright and dark spots of various shapes, distributed randomly throughout space.
- It arises from the interference of coherent light waves scattered by microscopically small irregularities on a surface.
- As these scattered waves overlap and interfere with each other it produces constructive and destructive interference

2. Explain the term (i) Induces Absorption (ii) Spontaneous emission (iii) stimulated emission

Sol:

(i) Induced absorption

- An atom has an infinite number of quantized energy states.
- *Let the atom be initially in the lower state E_1. If a photon of energy hν is incident on the atom in the lower state, the atom absorbs the incident photon and gets excited to the higher energy state E_2. This process is called induced absorption as shown in Figure.*
- For absorption: Atom + Photon → Atom*.

- If E_1 and E_2 are the energies of an electron in the initial and final states respectively and f is the frequency of absorbed radiation, then $E_2 - E_1 = h\nu$ or $\nu = (E_2 - E_1)/h$, where h is Planck's constant.
- The rate of absorption is proportional to the photon density (ρ) and to the number of atoms per unit volume N_1 in the ground state.

$$\text{Thus } R_{12} \propto N_1\rho \text{ OR}$$
$$R_{12} = B_{12}N_1\rho,$$

- Where $\mathbf{B_{12}}$ is called the (probability of absorption per unit time) **Einstein's coefficient for absorption of radiation**.

(ii) Spontaneous emission (Natural emission)

- *It is a process in which there is an emission of a photon whenever an atom transits from a higher energy state to a lower energy state without the aid of any external energy.*

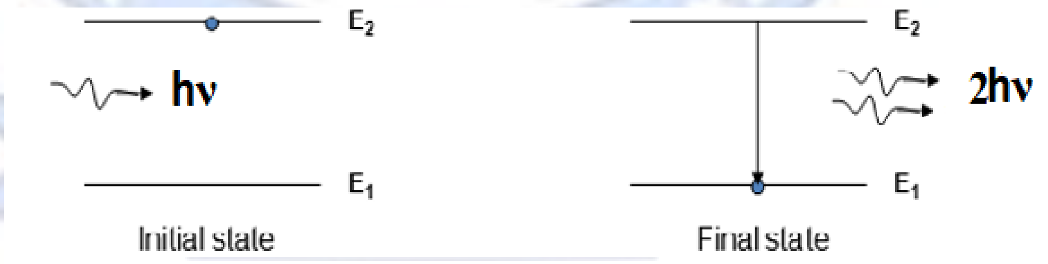
For spontaneous emission: Atom* → Atom + Photon.

- If E_1 and E_2 are the energies of an electron in the initial and final states respectively and the frequency of emitted photons, then $E_2 - E_1 = h\nu$ or $\nu = (E_2 - E_1)/h$, where h is Planck's constant.
- The rate of spontaneous emission is proportional to the number of atoms per unit volume in the excited state,

  i.e. $R_{21(sp)} \propto N_2$ or $R_{21(sp)} = A_{21}N_2$,
- Where $\mathbf{A_{21}}$ is called the (probability of spontaneous emission per unit time) **Einstein's coefficient of spontaneous emission of radiation.**

(iii) Stimulated emission (Forced emission)

- *It is a process in which there is an emission of a photon whenever an atom transits from a higher energy state to a lower energy state under the influence of an external energy.*
- (When the atom is in the excited state, then an incident photon of correct energy may cause the atom to transit to lower energy state, emitting an additional photon of the same frequency. Thus, now two photons of the same frequency are present.)
- For stimulated emission: Atom* + Photon → Atom + Photon + Photon



- If E_1 and E_2 are the energies of an electron in the initial and final states respectively and f the frequency of emitted photons, then $E_2 - E_1 = h\nu$ or $\nu = (E_2 - E_1)/h$, where h is Planck's constant.
- The rate of stimulated emission depends both on the external radiation and on the number of atoms per unit volume in the excited state E_2,

  i.e. $R_{21}(st) \propto N_2\rho$ or $R_{21}(st) = B_{21}N_2\rho$,
- Where $\mathbf{B_{21}}$ is called the (probability of stimulated emission per unit time) **Einstein's coefficient of stimulated emission of radiation**.

3. **Compare spontaneous emission and stimulated emission. (Mention four points of comparison)**

Sol:

Sr. No.	Spontaneous Emission	Stimulated Emission
1.	The emission of a light photon takes place immediately without any inducement.	The emission of a light photon takes place through an inducement. i.e. by an external photon.
2.	$R_{21(sp)} = A_{21}N_2$	$R_{21(st)} = B_{21}N_2\rho$
3.	It is natural emission	It is forced emission
4.	It is a random process	It is not a random process
5.	Emission takes place in all directions	Emission takes place in specific direction
6.	Intensity is low	Intensity is high
7.	Polychromatic light is emitted	Monochromatic light is emitted
8.	Photons will not be multiplied by chainreaction	Photons will be multiplied by chain reaction
9.	It is an uncontrollable process.	It is controllable process.

4. **Establish the relation between Einstein's coefficients.**

Or

Derive the relationship between Einstein Coefficients.

Sol:

Let us assume that the atomic system is in equilibrium with electromagnetic radiation. Hence at thermal equilibrium, the number of upward transitions is equal to the number of downward transitions per unit volume per second.

i.e., The rate of absorption = The rate of emission

$$B_{12} N_1 \rho = A_{21} N_2 + B_{21} N_2 \rho \qquad ...(4)$$

From the above equation,

$$(B_{12} N_1 - B_{21} N_2)\,\rho = A_{21} N_2$$

$$\rho = \frac{A_{21} N_2}{(B_{12} N_1 - B_{21} N_2)}$$

or

$$\rho = \frac{A_{21}}{\left(B_{12}\dfrac{N_1}{N_2} - B_{21}\right)} \qquad \text{[on dividing by } N_2\text{]} \qquad ...(5)$$

Under thermal equilibrium, the number of atoms N_1 and N_2 in the energy states E_1 and E_2 at a temperature T is given by Boltzmann distribution law. Hence, we have

$$N_1 = N_0\, e^{\left(\frac{-E_1}{k_B T}\right)} \qquad ...(6)$$

$$N_2 = N_0\, e^{\left(\frac{-E_2}{k_B T}\right)} \qquad ...(7)$$

where N_0 is the total number of atoms and k_B is the Boltzmann's constant.

$$\therefore \quad \frac{N_2}{N_1} = \frac{e^{\left(\frac{-E_2}{k_B T}\right)}}{e^{\left(\frac{-E_1}{k_B T}\right)}} = e^{\frac{-(E_2 - E_1)}{k_B T}}$$

$$\frac{N_2}{N_1} = e^{\left(\frac{-h\nu}{k_B T}\right)} \qquad [\text{Since } h\nu = E_2 - E_1]$$

or

$$\frac{N_1}{N_2} = e^{\left(\frac{h\nu}{k_B T}\right)} \qquad ...(8)$$

Substituting equation (8) in equation (5)

$$\rho = \frac{A_{21}}{\left[B_{12}\, e^{\left(\frac{h\nu}{k_B T}\right)} - B_{21}\right]} \qquad ...(9)$$

or

$$\rho = \frac{A_{21}}{B_{21}} \frac{1}{\left[\left(\frac{B_{12}}{B_{21}}\right)e^{\left(\frac{h\nu}{k_B T}\right)} - 1\right]} \qquad ...(10)$$

But from Planck's black body theory of radiation

$$\rho = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{\left(\frac{h\nu}{k_B T}\right)} - 1} \qquad ...(11)$$

Hence, comparing equations (10) and (11), we get

$$\frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3} \qquad ...(12)$$

$$B_{12} = B_{21} \qquad ...(13)$$

Take $B_{12} = B_{21} = B$ and $A_{21} = A$. ***The constants A and B are called Einstein's coefficients.***

Equation (13) shows that the probability of absorption is equal to the probability of stimulated emission.

From equation (12) it is seen that the ratio of spontaneous emission and stimulated emission is proportional to ν^3. It means that the probability of spontaneous emission dominates over stimulated emission.

5. **1. Explain the term (i) population inversion, (ii) pumping (iii) Lasing (iv)Active medium And (v) Lifetime (vi) metastable state (vii) Optical resonator.**

Sol:

- **BASICS DEFINITIONS (BASIC COMPONENTS):**

1. Population inversion

- Population inversion is a state of achieving more atoms in the excited state compared to those in ground state. i.e. $N_2 > N_1$.
- If this condition is satisfied, then there is more chance for stimulated emission to take place.
- Population inversion is an essential condition for producing laser.
- Population inversion can be achieved by a process called pumping.

2. Pumping

- Pumping is the process of exciting atoms from a lower energy state to a higher energy state by supplying energy from an external source.
- The most commonly used pumping processes are:

2.1 Optical pumping:

- In this type of pumping atoms are excited using an external optical (light) source.
- This type of pumping is used in solid-state lasers such as ruby laser and Nd: YAG laser.

2.2 Electrical pumping (Direct electron excitation):

- In this type of pumping atoms are excited using a high external electric field. The electrons are accelerated to a high velocity by a strong electric field with help of a discharge tube. These moving electrons collide with the natural gas atoms and ionize the medium.
- Thus, due to ionization they transit to a higher energy level. This type of pumping is used in gas laser such as CO_2 laser.

2.3 Direct conversion:

- In this type of pumping the direct conversion of electric energy into light takes place. This type of pumping is used in semiconductor lasers.

- Pumping by collision and chemical methods: In addition to the above three, the other types of pumping are, inelastic collision between atoms and chemical methods which are respectively adopted in He-Ne gas laser and dye and chemical lasers.

3. Lasing

The process that leads to the emission of stimulated photons after achieving population inversion is called lasing. (OR the process that helps to get stimulated photons after achieving population inversion is called lasing.)

4. Active medium

The medium in which population inversion is achieved for laser action is called the active medium.

The medium can be solid, liquid gas, and plasma.

Based on the active medium and method of pumping, the lasers are classified into:

1. Solid-state lasers
2. Liquid lasers
3. Gaseous lasers
4. Dye lasers
5. Semiconductor lasers

5. Lifetime

The time interval for which an atom can stay in a higher energy level is called lifetime (**10^{-9} s**).

6. Metastable state

The higher energy levels in which an atom can stay for a longer time (**10^{-3} to 10^{-2} s**) are called metastable states. This property helps in achieving population inversion.

7. Optical resonator (Resonator cavity)

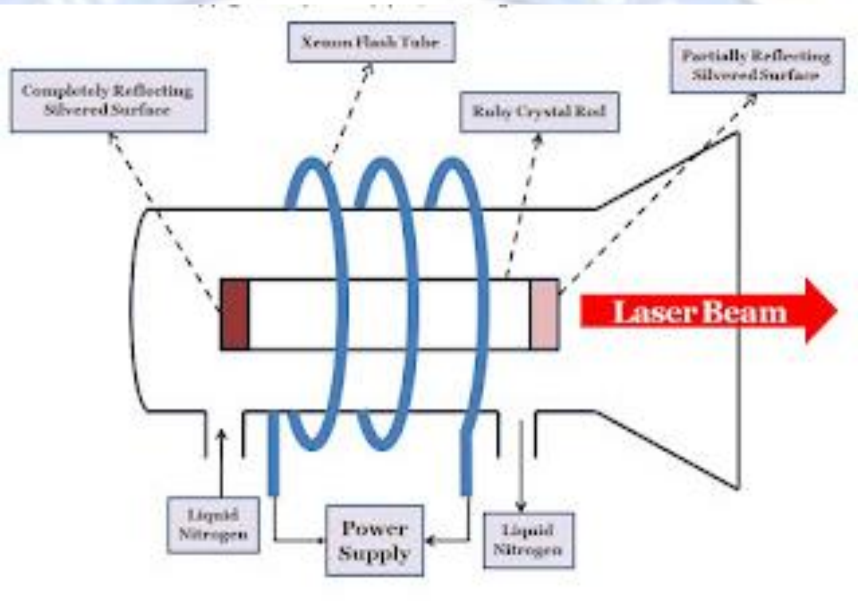
It is a **pair of reflecting mirrors** (surfaces) of which one is a perfect reflector and other is a partial reflector. It is used for amplification of photons.

6. Explain construction and working of Ruby LASER.
Or
Explain in detail construction and working of Ruby LASER with necessary schematic and energy level diagrams.

Sol:

- Ruby laser is the first successful laser developed by Theodore Maiman in 1960.
- Ruby laser is one of the few solid-state lasers that produce visible light. (deep red light of wavelength 694.3 nm)
- It is 3 level solid-state laser. It is a pulsed laser.

1 Construction



1.1 Active Medium:

A single crystal of ruby ($Al_2O_3:Cr^{+3}$) in form of cylinder (4cm long and 5mm diameter) acts as laser medium or active medium. Ruby is made of host sapphire (Al_2O_3) which is doped with small amount (0.05%) of chromium ions (Cr^{+3}). Chromium ions are active centers.

1.2 Pumping Source:

A helical Flash tube filled with xenon flash lamp is used as a pumping source.

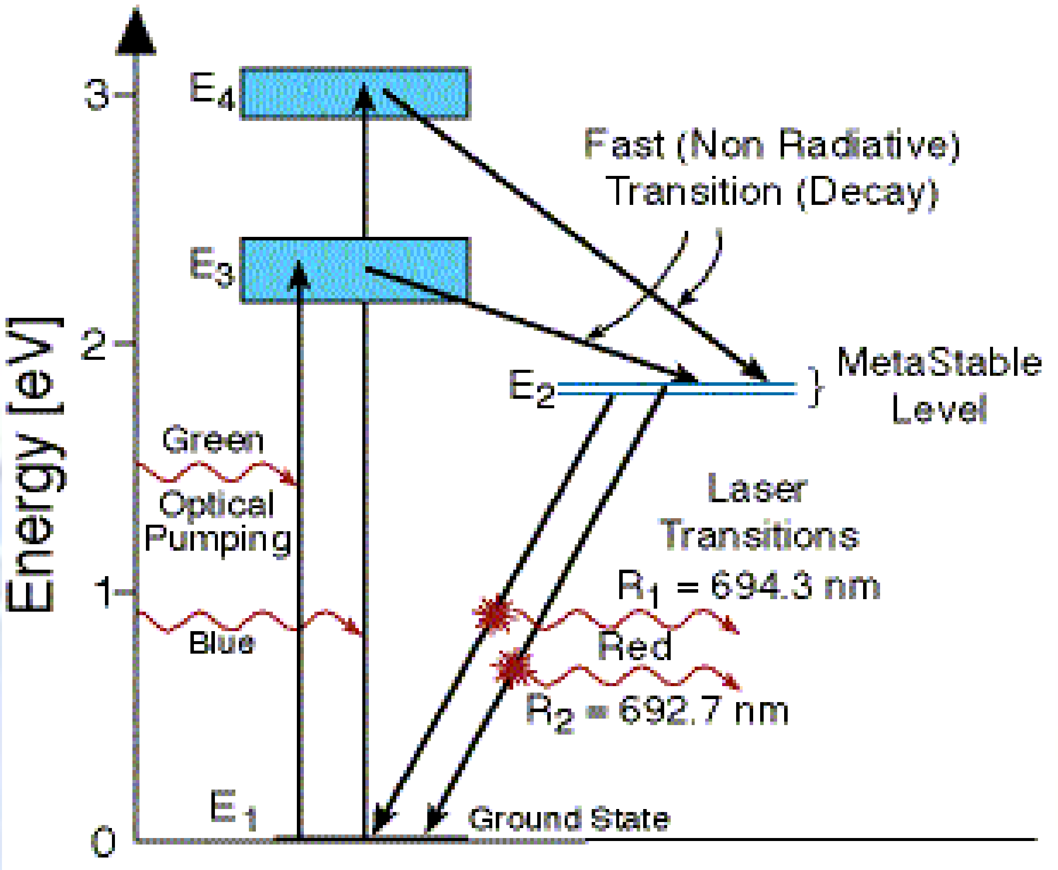
1.3 Resonator cavity (Optical resonator):

A pair of mirrors of which one is total reflector and the other is a partial reflector is called an optical resonator. (OR the end faces of the RUBY rod are ground polished and silvered to act as the optical resonator mirrors.)

1.4 Reflective cylinder:

The reflective cylinder is used to focus light on Ruby.

2. Working



- When the flash tube is switched on, the light will get focused on the rod, and chromium ions absorb energy of Green (5500 Å) and Blue (4200 Å) light and transit to higher energy levels E_3 and E_4.
- Transition of chromium ions from E_3 to E_2 and E_4 to E_2 is not radiative transition.
- E_2 behaves as metastable state. Population inversion can be achieved here.
- Transition of chromium ions from E_2 to E_1 due to external photons, gives stimulated emission. These stimulated photons after reflection from two mirrors produce an intense laser beam of wavelength 6943Å.
- Only a part of the energy emitted by the flash lamp is used to excite the Cr^{3+} ions, while the rest heats the crystal. Thus, the system can be cooled by liquid nitrogen circulation.

3. Applications

- Used in laboratory experiments.
- Used for soldering and welding.
- Used to drill fine holes in brittle materials
- Used in interferometry, holography.
- Used in Light Detection and Ranging (LIDAR).

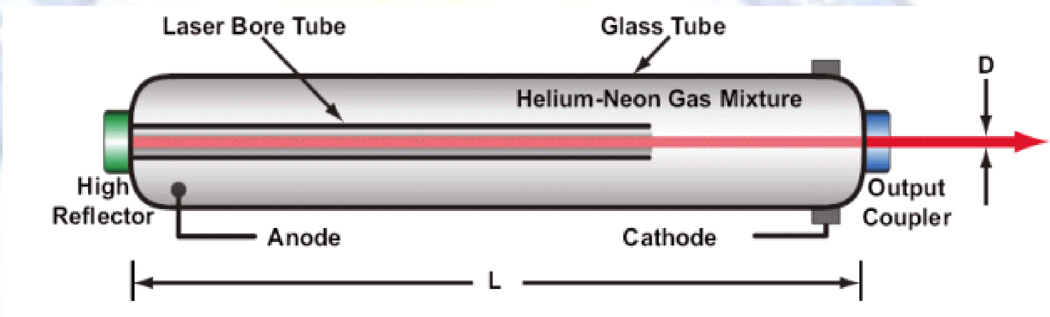
7. Explain in detail construction and working of He-Ne Gas LASER with necessary schematic and energy level diagrams.

Sol:

HELIUM-NEON GAS LASER (GASEOUS LASER)

- The first gas laser, the Helium-neon laser (He-Ne), was co-invented by Iranian-American physicist Ali Javan and American physicist William R. Bennett, Jr. in 1960.
- He-Ne laser is a four-level laser. Its usual operation wavelength is 632.8 nm, in the red portion of the visible spectrum. It operates in Continuous Working (CW) mode.
- A helium-neon laser, usually called a He-Ne laser, is a type of small gas laser. He-Ne lasers have many industrial and scientific uses and are often used in laboratory demonstrations of optics.

- ### Construction



1. Active Medium:

The active medium of the laser, as suggested by its name, is a mixture of helium and neon gases, in a 5:1 to 20:1 ratio, contained at low pressure (an average of 50 Pa per cm of cavity length) in a glass envelope. Generally, it is taken as 10:1.

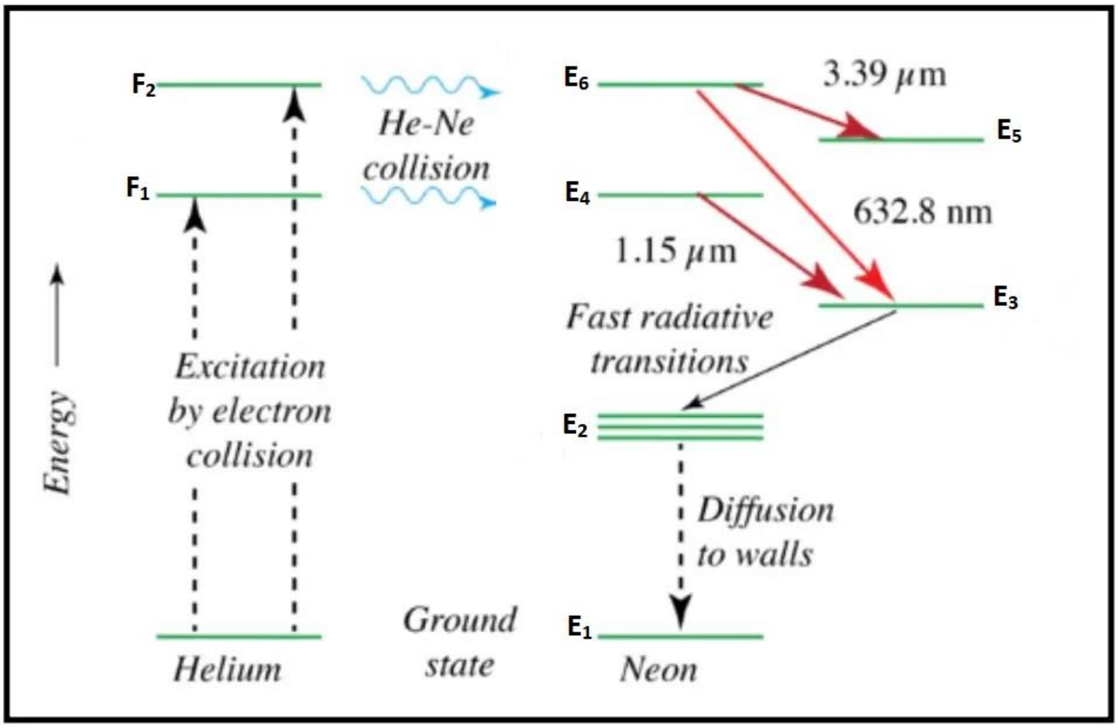
2. Pumping Source:

A discharge tube of length 80.6cm and bore diameter of 11.5cm.
The energy or pump source of the laser is provided by an electrical discharge of around 1000 volts through an anode and cathode at each end of the glass tube.
A current of 5 to 100 mA is typical for CW operation.

3. Resonator cavity (Optical resonator):

A pair of mirrors of which one is total reflector and other is partial reflector is called an optical resonator.

- **Working**



- The discharge tube is filled with Helium at a pressure of 1 mm of Hg & Neon at 0.1mm of Hg.
- When an electric discharge is set up in the tube, the electrons present in the electric field make collisions with the ground state He atoms.
- Hence ground state He atoms get excited to the higher energy levels F_1, F_2 .
- Here Ne atoms are active centers.
- The excited He atoms make a collision with the ground state Ne atoms and bring the Ne atoms into the excited states E_4 & E_6.
- The energy levels E_4 & E_6 of Ne are the metastable states and the Ne atoms are directly pumped into these energy levels.
- Since the Ne atoms are excited directly into the levels E_4 & E_6, these energy levels are more populated than the lower energy levels E_3 & E_5.
- Therefore, the population inversion is achieved between E_6 & E_5, E_6 & E_3, E_4 & E_3
- The transition between these levels produces wavelengths of 33912 A^0, 6328 A^0, and 11523 A^0 respectively.
- Now the Ne atoms undergo a transition from E_3 to E_2 and E_5 to E_2 in the form of fast decay giving photons by spontaneous emission. These photons are absorbed by optical elements placed inside the laser system.
- The Ne atoms are returned to the ground state (E_1) from E_2 by non-radiative diffusion and collision process, therefore, there is no emission of radiation.
- Some optical elements placed inside the laser system are used to absorb the IR laser wavelengths 33912 A^0, 11523 A^0.
- Hence the output of He-Ne laser contains only a single wavelength of 6328A^0.
- The released photons are transmitted through the concave mirror M_2 thereby producing a laser.

- A continuous laser beam of red colour at a wavelength of $6328A^0$.
- By the application of large potential difference, Ne atoms are pumped into higher energy levels continuously.
- A Laser beam of power 0.5 to 50 MW comes out from He-Ne laser.

- **Applications**
  - The Narrow red beam of the He-Ne laser is used in supermarkets to read bar codes.
  - The He- Ne Laser is used in Holography to produce 3D images of objects.
  - He-Ne lasers have many industrial and scientific uses and are often used in laboratory demonstrations of optics.
  - Low-level He-Ne laser is commonly used in therapy and has the advantage of having beneficial effects on tissue healing and pain relief.

8. Give applications of LASER in various fields in detail.

Sol:

1 In Industries

- For welding and melting.
- For cutting and drilling holes.
- To test the quality of the material.
- For the heat treatment of metallic and non-metallic materials.

2 In Medicine

- For removing eye defects such as myopia. For treatment of detached retina.
- In performing micro and bloodless surgery.
- For the treatment of human and animal cancers and skin tumors.

3 Military application

- For targeting.
- The laser beam can serve as a war weapon, i.e. a powerful laser beam can be used to destroy in a few seconds, big-sized objects like airplanes, missiles, etc., by pointing the laser beam onto them. For this reason, it can be even called a death ray.
- The laser beam can be used to determine precisely the distance, velocity, and directions as well as the size and form of distant objects of the reflected signal. It is known as LIDAR (Light Detection and Ranging).

4 Science and engineering applications

- In fiber optic communication.
- Communication between planets is possible with a laser.
- In holography.
- In underwater communication between submarines, as they are not easily absorbed by water.
- To accelerate some chemical reactions.
- To create new chemical compounds by destroying atomic bonds between molecules.
- To drill minute holes in the cell wall without damaging the cell.